CNC MACHINE OPERATIONS & CARE

CNC Error Code Reference

Example alarm categories • meanings • alarm-handling procedures

DOCUMENT CONTROL	DETAIL
Document	CNC_Error_Code_Reference.pdf
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Applicability	General CNC machining centres, lathes and routers; adapt locally

Important: This is an educational, generic reference. CNC machines differ substantially by builder, model, control, options, tooling and site rules. The machine-specific operator, maintenance, electrical and safety manuals take precedence. Do not use example values, procedures or alarm descriptions as substitutes for approved machine documentation.

1. How to use this reference

CNC alarm numbers and meanings are control- and machine-builder-specific. The examples below are **illustrative alarm categories**, not universal codes. A code such as “100”, “SV0401” or “EXxxx” can mean different things on different controls. Always consult the exact control model’s alarm manual and the machine builder’s alarm list.

Never infer a repair from an alarm label alone. Capture the complete message, subcode, axis/channel, timestamp and operating context. Do not repeatedly reset, defeat an interlock or change parameters to suppress an alarm.

2. Example alarm categories and first response

Example label (not universal)	Typical meaning / category	Initial operator response
EMERGENCY STOP ACTIVE	Emergency stop circuit is active or not satisfied.	Stop. Identify the activated E-stop and follow site reset procedure only after the hazard is cleared. If it will not reset, call maintenance.
DOOR / GUARD INTERLOCK	Guard or door safety condition not satisfied.	Keep guards closed. Check for obstruction from outside the guarded area. Never bridge or defeat the switch.
SERVO / AXIS ALARM	Axis drive reports a fault, overload or feedback issue.	Record axis and full drive code. Stop motion. Do not alter servo parameters; maintenance diagnoses.
OVERTRAVEL / LIMIT	Axis reached a software or hardware travel limit.	Do not force jog or defeat limit. Use only the documented recovery procedure with a clear area and authorized operator.
SPINDLE DRIVE FAULT	Spindle drive, motor, feedback, temperature or load fault.	Stop spindle and record speed/load context. Allow safe stop; maintenance checks drive and cooling.
SPINDLE OR TOOL CLAMP	Tool clamping/unclamping state not confirmed.	Do not run spindle. Keep clear of tool interface; follow OEM recovery and inspection procedure.
ATC / TOOL CHANGE TIMEOUT	Automatic tool changer did not complete a commanded step.	Do not reach into magazine. Capture sequence state and use the machine-specific recovery instructions.
LOW LUBRICATION LEVEL / PRESSURE	Lubrication reservoir or pressure monitoring indicates a problem.	Stop if required by the machine instructions. Check indicator/reservoir from safe access; use specified lubricant only.
LOW COOLANT LEVEL / FLOW	Coolant tank, pump or flow monitoring fault.	Stop cutting if coolant is essential. Check level and visible filter/nozzle condition under approved safe procedure.
HYDRAULIC PRESSURE LOW	Hydraulic pressure below monitored threshold.	Stop if clamping or machine function may be compromised. Do not work on pressurized lines.
AIR PRESSURE LOW	Pneumatic supply below required threshold.	Check plant-air status and displayed gauge. Do not change regulator settings beyond approved range.
PROGRAM / FORMAT ERROR	Invalid command, syntax, parameter or unsupported function.	Record the program line and exact message. Check control dialect, postprocessor, units and modal commands before rerunning.
OVERLOAD / EXCESSIVE LOAD	Axis, spindle or process load exceeded a configured threshold.	Stop and inspect tool, material, workholding and cutting conditions. Do not increase limits to continue.
THERMAL / OVER-TEMPERATURE	Motor, drive, spindle, cabinet or coolant temperature high.	Stop as instructed; check cooling indicators and ambient conditions. Do not open energized cabinets.

Example label (not universal)	Typical meaning / category	Initial operator response
BATTERY / MEMORY WARNING	Backup battery, memory or data retention warning.	Notify maintenance promptly. Follow the control-specific backup and battery replacement procedure; do not remove batteries casually.

3. Alarm-handling procedure

1. Stop the process safely. If immediate danger exists, use the emergency response procedure.
2. Read and record the exact alarm text, number, subcode and any associated alarms. Photograph the screen if permitted.
3. Record machine model/serial, control model/software version, mode, program/block, axis, tool and spindle state.
4. Identify whether the alarm is informational, process-related, motion-related or safety-critical using the OEM alarm manual.
5. Perform only operator-level checks listed in the manual (e.g., utility status, guard closure, tool identity, coolant level).
6. If the cause is not clear or the alarm returns, stop and escalate to qualified maintenance or OEM service.
7. Reset only when the manual permits and the cause has been corrected. Use a controlled restart and verify safe operation.
8. Log the alarm, cause, corrective action, verification and any follow-up work order.

4. Alarm severity guide (site must define)

Suggested class	Examples	Response expectation
Critical / safety	E-stop circuit fault, guard/interlock fault, unexpected motion, electrical smoke, uncontrolled pressure release.	Stop; isolate/secure as required; do not restart until competent inspection and release.
Major / machine protection	Servo fault, spindle over-temperature, lubrication failure, hydraulic clamp pressure loss, repeated overload.	Stop the affected operation; capture evidence; maintenance assessment before restart.
Process / quality	Tool life warning, coolant concentration warning, dimensional trend, chip conveyor warning (where safe).	Follow documented operator response; assess part quality and correct condition before continuing.
Information	Message or reminder with no immediate hazard, as defined by OEM.	Acknowledge only as permitted; record if recurring and complete required action.

5. Example alarm log

Date/time	Code & exact text	Context / machine state	Action / reset	Outcome / owner
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

6. Data integrity and parameter protection

- Back up CNC programs, offsets, tool data, parameters and PLC data according to the builder’s approved schedule and site controls.
- Restrict parameter, ladder/PLC and servo tuning access to authorized personnel.

- Record the original value, approved change, reason, authorization and verification for every parameter change.
- Do not load parameter files from another machine unless the builder explicitly approves compatibility.
- After control replacement, battery service or memory warning, verify backups and machine configuration before production.

7. When to contact OEM service

Contact the builder or authorized service provider for repeated unexplained alarms, safety-circuit faults, encoder/feedback errors, spindle or axis drive faults, tool changer recovery beyond the operator procedure, parameter corruption, collision damage, or any condition requiring energized electrical diagnosis. Provide the complete alarm log and machine/control identification.